Experiment-16

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```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from collections import Counter

fruits =
["Apple","Banana","Mango","Apple","Orange","Mango","Banana","Apple","Ma
ngo","Orange",

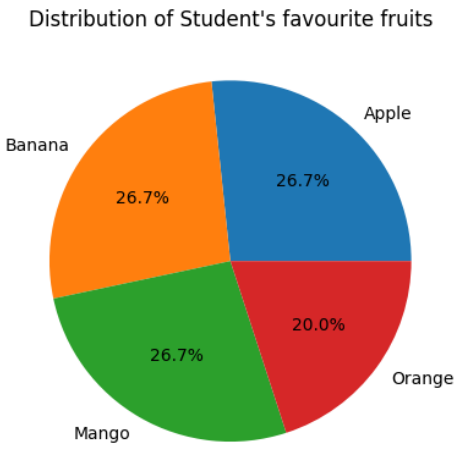
    "Banana","Mango","Apple","Orange","Banana"]

fruits_counts = Counter(fruits)

plt.pie(fruits_counts.values(), labels = fruits_counts.keys(),autopct =
'%1.1f%%')

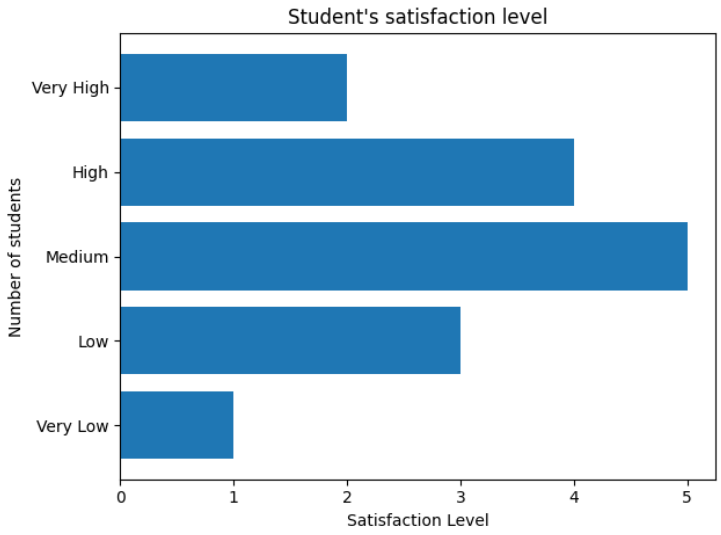
plt.title("Distribution of Student's favourite fruits")

plt.show()
```

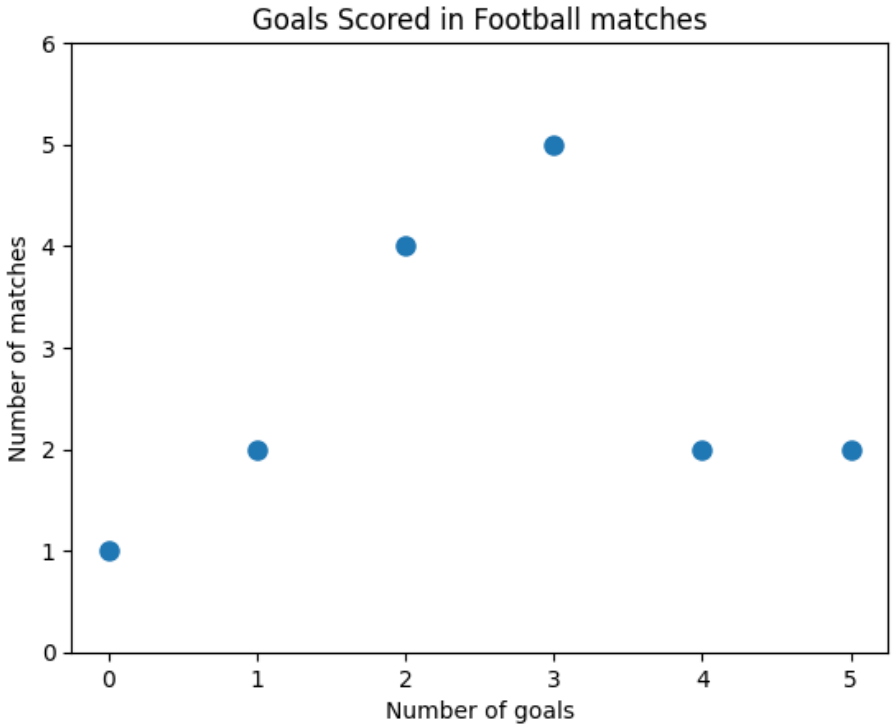


```
satisfaction = [
    "Low","Medium","High","Medium","Very High",
      "Low","High","Medium","Very High","High",
        "Low","Medium","High","Very Low","Medium"
        ]

category_order = ["Very Low","Low","Medium","High","Very High"]
count = Counter(satisfaction)
frequencies = [count[category] for category in category_order]
plt.barh(category_order,frequencies)
plt.title("Student's satisfaction level")
plt.xlabel("Satisfaction Level")
plt.ylabel("Number of students")
plt.tight_layout()
plt.show()
```



```
goal_count = [2,5,3,1,4,2,3,0,3,2,5,1,4,3,3,2]
frequency = Counter(goal_count)
values = sorted(frequency.keys())
frequencies = [frequency[value] for value in values]
plt.plot(values,frequencies,marker = 'o',linestyle = 'None',markersize = 8)
plt.title("Goals Scored in Football matches")
plt.xlabel("Number of goals")
plt.ylabel("Number of matches")
plt.xticks(range(0,6))
plt.yticks(range(0,7))
plt.show()
```



```
heights = [
    150.5,162.3,155.8,170.2,165.7,
      158.4,172.6,160.1,168.9,154.6,
        163.5,175.2,159.7,166.4,152.8
        ]
plt.hist(heights,bins = 5,edgecolor = 'black')
plt.title("Distribution of Student's Heights")
plt.xlabel("Height (cm)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

